The mixture which fills the entire cylinder volume is now compressed into the clearance volume. At the end of the compression stroke the mixture is ignited with the help of a spark plug located on the cylinder head.

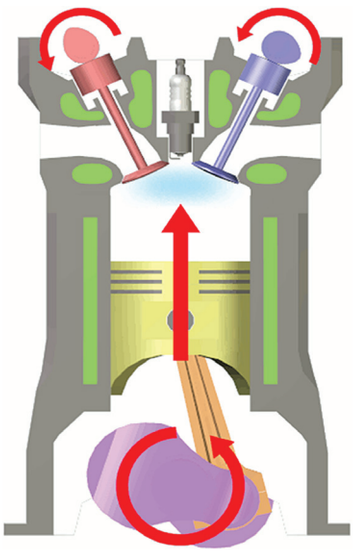


Figure 5.4(b)

c) Power Stroke

Rapid combustion of the fuel releases heat and there is an increases in the temperature inside the combustion chamber. The increased temperature of the gases also produces an increased pressure in the combustion chamber. The high pressure of the gases acting on the face of the piston causes the piston to move from TDC to BDC. This reciprocating motion is converted into the rotary motion of crankshaft through connecting rod. During power stroke, both inlet and exhaust valves are in closed condition. As the piston travels downward, there will be drop in combustion pressure and temperature as the volume increases. At the end of power stroke, exhaust stroke will be started.

d) Exhaust Stroke

At the end of power stroke, exhaust stroke will be started. During this stroke, the exhaust vavle will be open and the inlet value will be closed. When the piston from BDC to TDC, the burnt air fuel mixture is sent out through exhaust valve.

At the end of exhaust stroke, the intake stroke of the next cycle starts and this keep the engine in running condition.

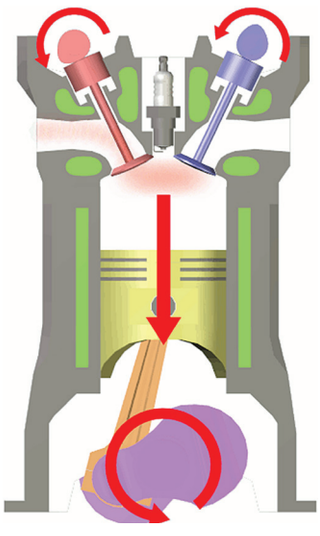


Figure 5.4(c)

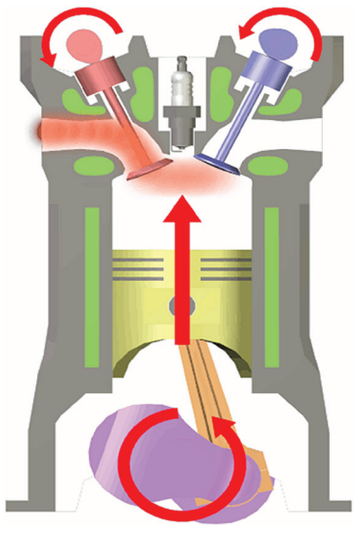


Figure 5.4(d)

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